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DEPARTMENT OF CSE/CS&IT

PROJECT ABSTRACT SUBMISSION

COURSE: Big Data Analytics (23DSB3303R) **A.Y. 2025 - 2026**

TITLE OF THE PROJECT	Financial Transaction Pattern Analysis Using Big Data Analytics	
S. No.	**University ID**	**Name**
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Abstract

The rapid growth of digital payment systems and online banking has led to the generation of massive volumes of financial transaction data, making effective analysis critical for understanding transaction behaviour, identifying meaningful patterns, and detecting suspicious activities. This project, Financial Transaction Pattern Analysis Using Big Data Analytics, focuses on the structured analysis of large, real-world, and unclean financial datasets. The dataset used in this study is obtained from a publicly available source and contains common data quality issues such as missing values, duplicate records, and inconsistent formats. To address these challenges, data preprocessing and cleaning techniques including data imputation, duplicate removal, normalization, and format standardization are applied to enhance data reliability and consistency. Following preprocessing, exploratory data analysis (EDA) and descriptive analytics are performed to examine transaction frequency, high-value transactions, and overall customer behaviour. Big Data Analytics concepts are employed to manage challenges related to data volume, variety, and veracity. Analytical models such as clustering techniques (e.g., K-means) for customer segmentation, statistical trend analysis, and rule-based and anomaly detection models are used to identify unusual transaction patterns. Additionally, classification and regression models support behaviour prediction and pattern identification. The outcomes of the analysis demonstrate how meaningful insights can be extracted from raw financial data, emphasizing the importance of data preprocessing and analytical modelling in enhancing monitoring, analysis, and decision-making within banking and financial institutions.